

The most useful restriction enzymes cleave the sugar-phosphate backbones in the two DNA strands in a staggered manner, as shown in step ❶ of Figure 20.5. The resulting double-stranded restriction fragments have at least one single-stranded end, called a **sticky end**. These short extensions can form hydrogen-bonded base pairs with complementary sticky ends on any other DNA molecules cut with the same restriction enzyme, such as the inserted DNA shown in step ❷ of Figure 20.5. The associations formed in this way are only temporary but can be made permanent by DNA ligase, an enzyme that catalyzes the formation of covalent bonds that close up the sugar-phosphate backbones of DNA strands (see step ❸ of Figure 20.5). At the bottom of Figure 20.5, you can see the stable recombinant DNA molecule that was produced by the ligase-catalyzed joining of DNA from two different sources. The end result, in this example, is the formation of a stable recombinant plasmid containing foreign DNA.

➔ **Mastering Biology Animation: Creating Recombinant DNA**

To check the recombinant plasmids after they have been copied many times in host cells to make sure the fragment has been inserted (see Figure 20.4), a researcher might cut the products again using the same restriction enzyme. If the insert is there, there would be two DNA fragments, one the size of the plasmid and one the size of the inserted DNA. To separate and visualize the fragments, researchers carry out a technique called **gel electrophoresis**, which uses a gel made of a polymer that has microscopic holes of different sizes, through which shorter fragments can travel faster. The gel works as a molecular sieve to separate out a mixture of nucleic acid fragments by length (Figure 20.6). Gel electrophoresis is used in conjunction with many different techniques in molecular biology.

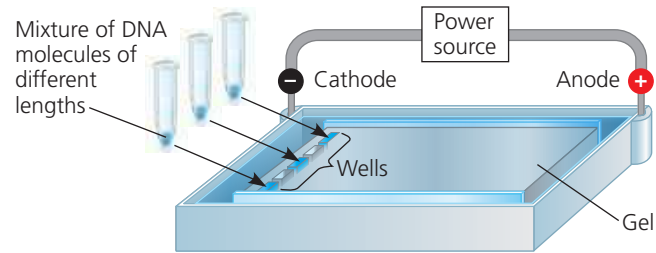
Amplifying DNA: The Polymerase Chain Reaction (PCR) and Its Use in DNA Cloning

Now that we have examined the cloning vector in some detail, let's consider how biologists obtain the foreign DNA to be inserted. Most researchers have some sequence information about the DNA fragment they want to clone. Using this information, they can start with genomic DNA from the particular species of interest and obtain many copies of the desired gene by using a technique called the **polymerase chain reaction**, or **PCR**. Figure 20.7 illustrates the steps in PCR. Within a few hours, this technique can make billions of copies of a specific target DNA segment in a sample, even if that segment makes up less than 0.001% of the total DNA in the sample.

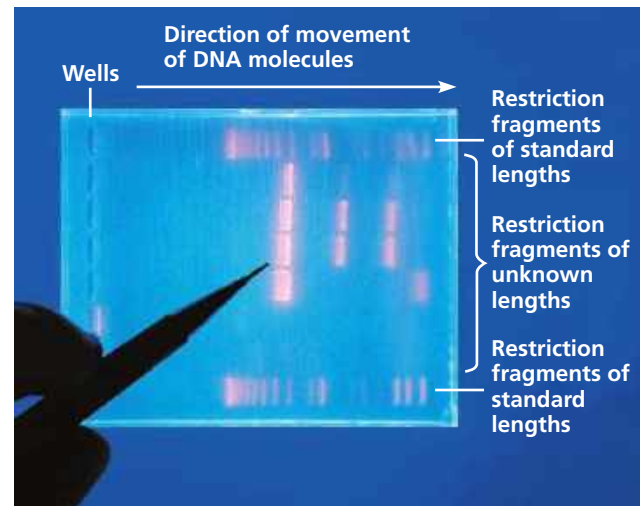
➔ **Mastering Biology HHMI Video: Polymerase Chain Reaction (PCR)**



▼ **Figure 20.6 Gel electrophoresis.** A gel made of a polymer acts as a molecular sieve to separate nucleic acids or proteins differing in size, electrical charge, or other physical properties as they move in an electric field. In the example shown here, DNA molecules are separated by length in a gel made of a polysaccharide called agarose.



(a) Each sample, a mixture of different DNA molecules, is placed in a separate well near one end of a thin slab of agarose gel. The gel is set into a small plastic support and immersed in an aqueous, buffered solution in a tray with electrodes at each end. The current is then turned on, causing the negatively charged DNA molecules to move toward the positive electrode.



(b) Shorter molecules are slowed down less than longer molecules, so shorter molecules move faster through the gel. After the current is turned off, a DNA-binding dye is added that fluoresces pink in ultraviolet (UV) light. Each pink band corresponds to many thousands of DNA molecules of the same length. The bands at the upper and lower edges of the gel are restriction fragments of standard lengths for comparison with samples of unknown length.

➔ **Mastering Biology Animation: Gel Electrophoresis of DNA**

In the PCR procedure, a three-step cycle causes a chain reaction that produces an exponentially growing population of identical DNA molecules. During each cycle, the reaction mixture is ❶ heated to high temperatures to denature (separate) the strands of the double-stranded DNA and then ❷ cooled to allow annealing (hydrogen bonding) of short, single-stranded DNA primers complementary to sequences on opposite strands at each end of the target sequence; finally, ❸ a special DNA polymerase extends the primers in the 5' → 3' direction. This cycle is then repeated 30–40 times. If a standard DNA polymerase were used, this enzyme